

Technical Analysis

Canny Thresholds and Hough Parameters

Parameter	Value	Rationale
τ_{low} (Canny)	50	Captures subtle lane edges while suppressing low-contrast noise
τ_{high} (Canny)	150	1:3 ratio with τ_{low} ; standard practice for edge detection
ρ (Hough)	1 px	Fine distance resolution for accurate lane positioning
θ (Hough)	$\pi/180$	1° angular resolution balances precision and computation
Threshold	30	Sufficient votes to confirm a line without being too strict
minLineLength	20 px	Filters out very short noise segments
maxLineGap	200 px	Bridges gaps in dashed lane lines

ML-Based Line Fitting - Mathematical Justification

- Simple Averaging** computes $\bar{m} = \frac{1}{N} \sum_{i=1}^N m_i$ and $\bar{b} = \frac{1}{N} \sum_{i=1}^N b_i$. This is the maximum likelihood estimator under a Gaussian noise assumption but is sensitive to outliers.
- RANSAC** iteratively:
 - Samples a minimal subset of (m, b) pairs
 - Fits a model to the subset
 - Counts inliers within a tolerance
 - Keeps the model with the most inliersThis makes it robust to outlier segments caused by shadows, road markings, or other artifacts.
- K-Means** (k=1) minimizes $\sum_{i=1}^N \|(m_i, b_i) - (\bar{m}, \bar{b})\|^2$, which for k=1 is mathematically equivalent to computing the centroid (i.e., the mean). However, it generalizes to k>1 for multi-lane detection.

Limitations

- Cluttered scenes (image5): Non-lane features (barriers, vehicles, road texture) generate edge segments that pass the slope filter, corrupting fitted lines
- Faint/absent markings (image1, image4): Roads with weak or missing markings on one side yield asymmetric detection (one lane only)
- Curved roads: The linear model $y = mx + b$ cannot capture lane curvature; polynomial fitting or sliding window approaches would be needed
- No semantic understanding: The pipeline treats all edges equally; it cannot distinguish a lane line from a barrier or vehicle edge based on appearance

Recommendations

- Use semantic segmentation (e.g., LaneNet, SCNN) for scenes with complex infrastructure
- Apply polynomial fitting instead of linear for curved roads
- Implement temporal smoothing across video frames for stability
- Consider adaptive Canny thresholding (e.g., Otsu-based) for automatic parameter selection
- Use narrower ROI or edge density thresholding to reduce false positives in cluttered scenes